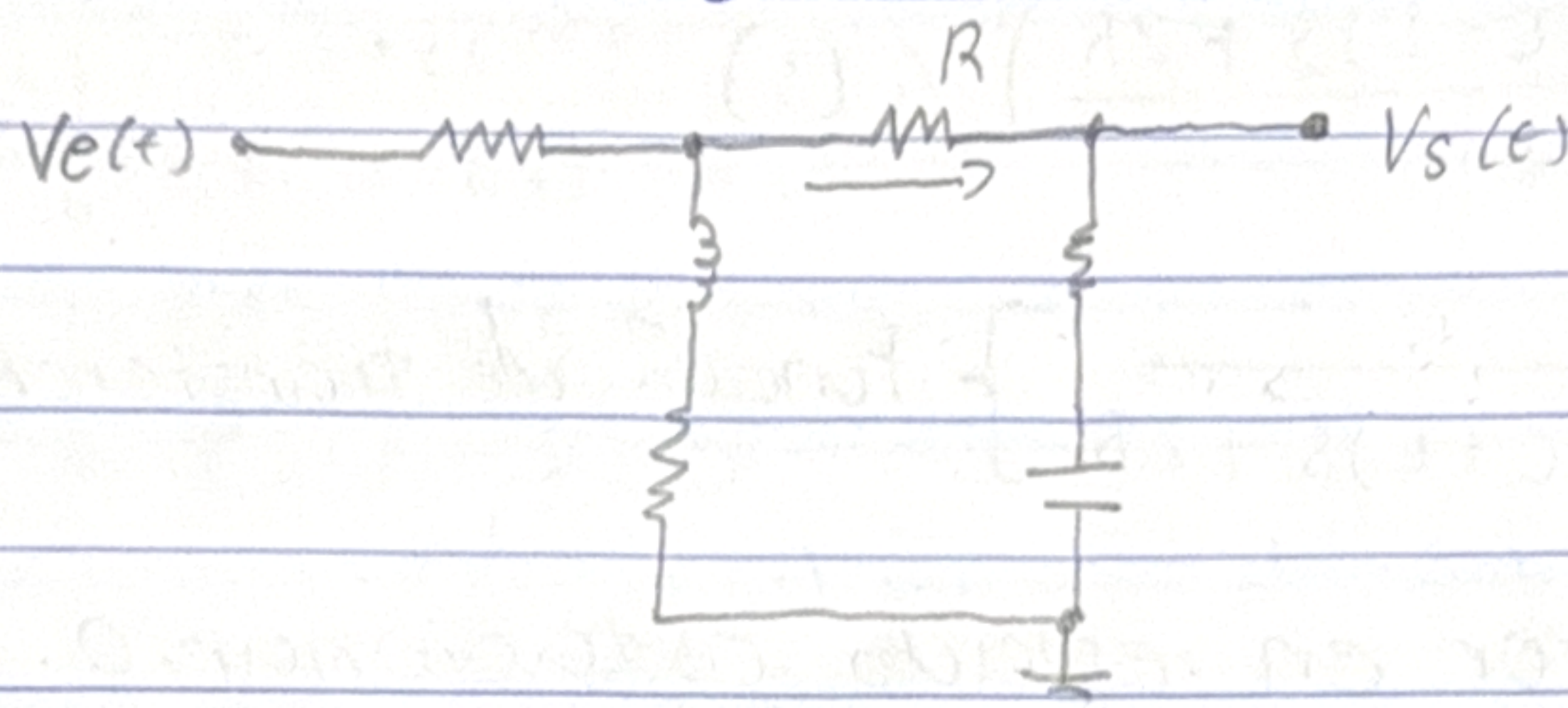


5.1 Practica: Diseño de controlador para un sistema de segundo orden.

Objetivo: Diseñar un controlador que permita eliminar el error entre la entrada $[V_e(t)]$ y la salida $[V_s(t)]$ del circuito RLC de la figura 5.1.



Diseño de un controlador para un sistema de segundo orden.

* Ecuaciones Principales:

Ecuación - Primera Malla... $V_e(t) = R i_1(t) + L \frac{d[i_1(t) - i_2(t)]}{dt} + R[i_1(t) - i_2(t)]$

Ec. - Segunda Malla...

$$L \frac{d[i_1(t) - i_2(t)]}{dt} + R[i_1(t) - i_2(t)] = R i_2(t) + R i_2(t) + \frac{1}{C} \int i_2(t) dt$$

$$V_s(t) = R i_2(t) + \frac{1}{C} \int i_2(t) dt$$

* Transformadas de Laplace

$$V_e(s) = R i_1(s) + L s [i_1(s) - i_2(s)] + R [i_1(s) - i_2(s)] \quad \text{Ec... 1}$$

$$L s [i_1(s) - i_2(s)] + R [i_1(s) - i_2(s)] = R i_2(s) + R i_2(s) + \frac{i_2(s)}{C s} \quad \text{Ec... 2}$$

$$V_s(s) = R i_2(s) + \frac{i_2(s)}{C s}$$

* Procedimiento algebraico para encontrar la función de transferencia

$$\text{Relación: } \frac{V_s(s)}{V_e(s)} = \frac{[\text{Polinomio}] \dot{I}_2(s)}{[\text{Polinomio}] \dot{I}_2(s)}$$

$$V_e(s) = (LS + 2R) \dot{I}_1(s) - (LS + R) \dot{I}_2(s)$$

$$(LS + R) \dot{I}_1(s) = (LS + \frac{1}{CS} + 3R) \dot{I}_2(s)$$

$$\dot{I}_1(s) = \left(\frac{1}{LS + R} \right) (LS + \frac{1}{CS} + 3R) \dot{I}_2(s)$$

$$V_s(s) = (R + \frac{1}{CS}) \dot{I}_2(s)$$

$$V_e(s) = (LS + 2R) \left(\frac{1}{LS + R} \right) (LS + \frac{1}{CS} + 3R) \dot{I}_2(s) - (LS + R) \dot{I}_2(s)$$

$$V_e(s) = (LS + 2R) \left(\frac{LCS^2 + 3RCS + 1}{CS(LS + R)} \right) \dot{I}_2(s) - (LS + R) \dot{I}_2(s)$$

$$V_e(s) = (LS + 2R) \left(\frac{LCS^2 + 3RCS + 1}{LCS^2 + RCS} \right) \dot{I}_2(s) - (LS + R) \dot{I}_2(s)$$

$$* V_e(s) = \left[(LS + 2R) \left(\frac{LCS^2 + 3RCS + 1}{LCS^2 + RCS} \right) - (LS + R) \right] \dot{I}_2(s)$$

$$* V_s(s) = (R + \frac{1}{CS}) \dot{I}_2(s) = \left(\frac{CRS + 1}{CS} \right) \dot{I}_2(s)$$

$$\frac{V_s(s)}{V_e(s)} = \frac{\left(\frac{CRS + 1}{CS} \right) \dot{I}_2(s)}{\left[(LS + 2R) \left(\frac{LCS^2 + 3RCS + 1}{LCS^2 + RCS} \right) - (LS + R) \right] \dot{I}_2(s)}$$

$$\frac{V_s(s)}{V_e(s)} = \frac{\frac{CRS + 1}{CS}}{\frac{L^2CS^3 + 3RLCS^2 + LS + 2LRCS^2 + 6R^2CS + 2R}{LCS^2 + RCS} - (LS + R)}$$

Por el prope:

$$\left\{ \frac{V_s(s)}{V_e(s)} = \frac{CLRS^2 + (CR^2 + L)S + R}{3CLRS^2 + (5CR^2 + L)S + 2R} \right\} \text{ Función de transferencia.}$$

* Error en estado estacionario

$$e(s) = 0.5 = \frac{1}{2} \quad e(s) = \lim_{s \rightarrow 0} s \cdot V_e(s) \left(1 - \frac{V_g(s)}{V_e(s)}\right) = \frac{1}{s-70} \cdot \frac{1}{s} \left(1 - \frac{R}{2R}\right)$$

$$e(t) = 0.5V = \frac{1}{2} \quad e(s) \left(1 - \frac{1}{2}\right) = 1\left(\frac{1}{2}\right) = \frac{1}{2} = 0.5$$

* Estabilidad en lazo abierto.

$$R = 2.2k\Omega$$

$$3CLR^2s^2 + (5CR^2 + L)s + 2R$$

$$L = 4.7mH$$

$$3.102 \times 10^{-4} s^2 + 242.0047 s + 4400$$

$$C = 10\mu f$$

$$X_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$a = 3.102 \times 10^{-4}$$

$$b = 242.0047$$

$$c = 4400$$

$$X_1 = \frac{-(242.0047) + \sqrt{(242.0047)^2 - (4)(3.102 \times 10^{-4})(4400)}}{(2)(3.102 \times 10^{-4})}$$

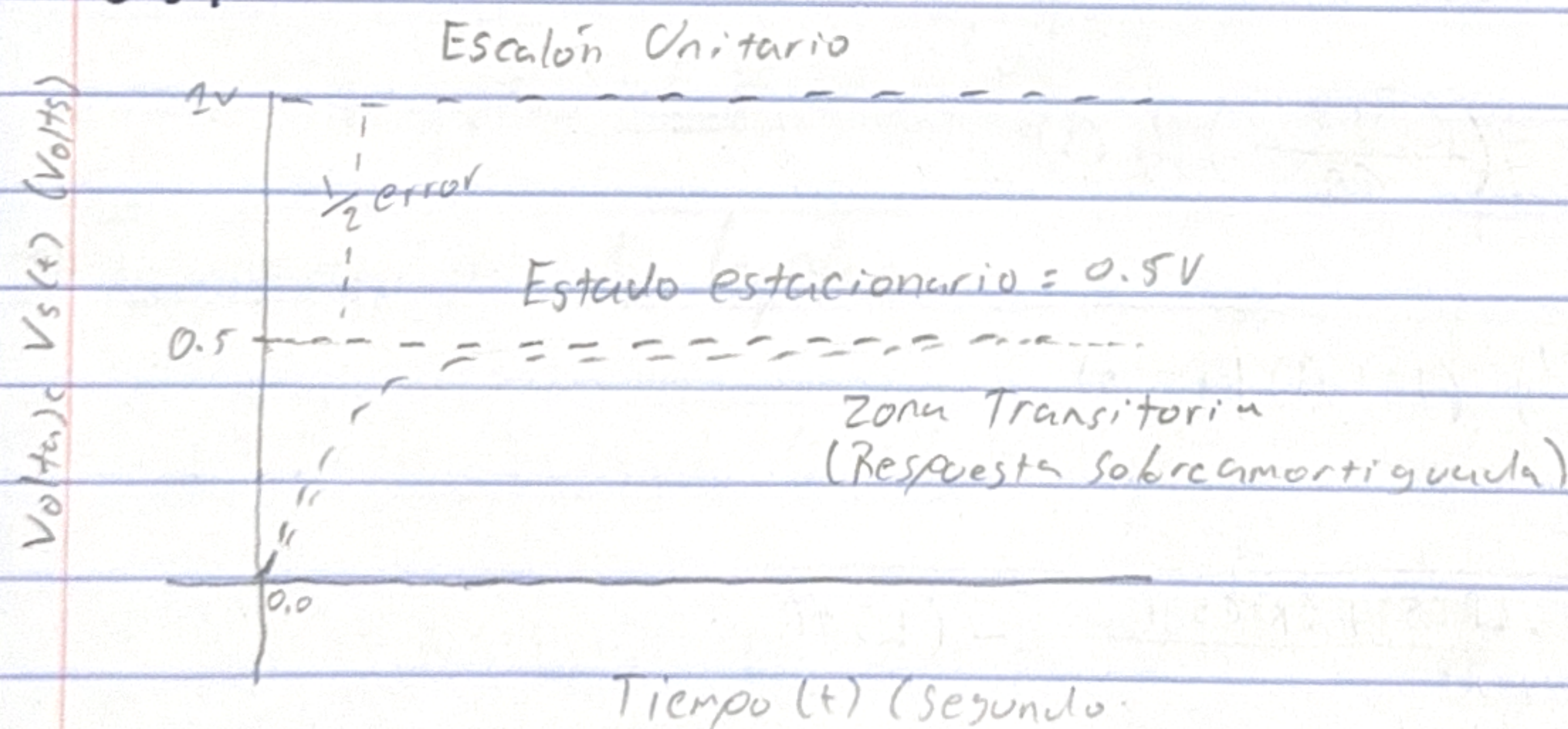
$$X_1 = -18.18188881$$

$$X_2 = \frac{-(242.0047) - \sqrt{(242.0047)^2 - (4)(3.102 \times 10^{-4})(4400)}}{(2)(3.102 \times 10^{-4})}$$

$$X_2 = -780138.8136$$

Polos negativo Reales? Sobreamortiguado. (estable)

* Bosquejo:



X Modelo de ecuaciones integro-diferenciales

El modelo se expresa despejando las variables dependientes, es decir, las corrientes:

$$V_c(t) = 2Ri_1(t) + L \frac{d[i_1(t) - i_2(t)]}{dt} - Ri_2(t)$$

$$i_1(t) = [V_c(t) + Ri_2(t) - L \frac{d[i_1(t) - i_2(t)]}{dt}] \frac{1}{2R}$$

$$L \frac{d[i_1(t) - i_2(t)]}{dt} + Ri_1(t) = Ri_2(t) + Ri_2(t) + Ri_2(t) + \frac{1}{C} \int i_2(t) dt$$

$$i_2(t) = [L \frac{d[i_1(t) - i_2(t)]}{dt} + Ri_1(t) - \frac{1}{C} \int i_2(t) dt] \frac{1}{3R}$$

$$i_1(t) = [V_c(t) + Ri_2(t) - L \frac{d[i_1(t) - i_2(t)]}{dt}] \frac{1}{2R}$$

$$i_2(t) = [L \frac{d[i_1(t) - i_2(t)]}{dt} + Ri_1(t) - \frac{1}{C} \int i_2(t) dt] \frac{1}{3R}$$

$$V_s(t) = Ri_2(t) + \frac{1}{C} \int i_2(t) dt$$

$$\frac{V_s(s)}{V_c(s)} = \frac{(10 \mu F)(4.7 mH)(2.2 k\Omega) s^2 + ((10 \mu F)(2.2 k\Omega)^2 + 4.7 mH) s + 2.2 k\Omega}{(3)(10 \mu F)(4.7 mH)(2.2 k\Omega) s^2 + (5)(10 \mu F)(2.2 k\Omega)^2 + 4.7 mH) s + 2(2.2 k\Omega)}$$

$$\frac{V_s(s)}{V_c(s)} = \frac{1.034 \times 10^{-4} s^2 + 48.4047 s + 2200}{3.102 \times 10^{-4} s^2 + 242.0047 s + 4400}$$